

Detection of SQL Antipatterns in jOOQ Database Access Code Using Large Language Models

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Background

- **SQL Antipatterns**
 - "... a commonly occurring solution to a problem that generates decidedly negative consequences" (Brown et al., 1998)
 - Can affect performance, maintainability, portability, and data integrity
 - **Example:** *Implicit Columns* (using `SELECT *`) can waste >25% of runtime/energy
- **jOOQ:** Popular Java library for type-safe, DSL-based database access

SQL Antipatterns in Java: literature review

- Prevalent in Java projects
 - *Implicit Columns* affects every 50th query
 - Remain unfixed longer than traditional code smells
 - Authors suspect lack of awareness or low priority
- Two tools capable of statically detecting SQL antipatterns in Java
 - None capable of detecting from jOOQ DSL

Research Goals

- Development of an LLM-based static analysis tool for detecting SQL antipatterns
- Studying the prevalence and occurrence patterns of SQL antipatterns in jOOQ

Research Questions

- **RQ1:** How do different LLMs compare in identifying SQL antipatterns in Java code that uses jOOQ for database access?
- **RQ2:** How do different prompting strategies affect the detection performance of LLMs in identifying SQL antipatterns in Java code that uses jOOQ for database access?
- **RQ3:** How accurately can the developed LLM-based tool detect SQL antipatterns in Java code that uses jOOQ for database access?
- **RQ4:** What patterns can be observed in the occurrence of SQL antipatterns in Java code that uses jOOQ for database access?
- **RQ5:** Which jOOQ API methods are most frequently associated with query antipattern occurrences?

Dataset Creation

- Used GitHub API to mine for projects using jOOQ
 - Filtered out non-Java projects, duplicates, etc
 - Gathered **602** applicable projects
- Annotated a subset of 10% projects
 - Stratified sampling using head-tail breaks to preserve size distribution
 - 4 large, 10 medium, 47 small projects
 - **1,562** total antipattern occurrences identified manually

Dataset Creation

- Internal consistency measured using Cohen's Kappa
 - Re-annotated 10% of files after one month

$$\kappa = \frac{p_o - p_e}{1 - p_e} = 0.834$$

- Result: *Almost perfect agreement*
- **61** projects split into training/validation/test sets
 - Split ratio of 34/33/33
 - Monte Carlo optimisation to achieve an optimal split

Analysis of LLMs and prompting strategies

- Models:
 - OpenAI GPT-5.2 — reasoning model
 - Anthropic Claude 4.5 Opus — non-reasoning* model
 - Z.ai GLM-5 — open model
 - OpenAI gpt-oss-120B — medium-sized open model
- Prompting strategies:
 - Zero-Shot — baseline
 - Few-Shot — providing examples
 - Chain-of-Thought — "Think step by step."
 - Tree-of-Thought — simulating a conversation between three experts

Analysis of LLMs and prompting strategies

- Designed prompts for each strategy, evaluating against training set
- Evaluated performance of each LLM & strategy combination on validation set

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = 2 \cdot \frac{P \cdot R}{P + R}$$

- Localisation task: True Positive only if model found the correct lines according to Intersection over Union

$$\text{IoU} = \frac{|S_p \cap S_g|}{|S_p \cup S_g|} \geq 0.5$$

RQ1: How do different LLMs compare in identifying SQL antipatterns in Java code that uses jOOQ for database access?

- All large models demonstrated similar performance
 - Varying costs and runtimes
 - GPT-5.2 was more paranoid than others
- gpt-oss-120B suffered from anomalies degrading performance

Model	Precision	Recall	F1-Score	Cost (USD)	Runtime (s)
GPT-5.2 (reasoning)	0.85	0.92	0.88	26.16	2,423
GLM-5 (reasoning)	0.88	0.89	0.88	11.11	5,775
Claude Opus 4.5 (non-reasoning)	0.88	0.89	0.88	29.97	367
gpt-oss-120B (reasoning)	0.83	0.87	0.83	1.49	2,808

RQ2: How do different prompting strategies affect the detection performance of LLMs in identifying SQL antipatterns in Java code that uses jOOQ for database access?

- 2% performance gains to non-reasoning model from CoT and ToT
- Up to 1% gains to open models from Few-Shot
- Gains for one model generally balanced by degradations for other models
- Large increases to cost and runtime

Strategy	Avg F1-Score Increase	Avg Cost Increase	Avg Runtime Increase
Few-Shot	0.6%	31%	-1%
Chain-of-Thought	-0.3%	23%	90%
Tree-of-Thought	0.1%	88%	638%

Analysis tool development and evaluation

- **Stack:** Bun (build tool & runtime), TypeScript, Vercel AI SDK.
- **Architecture:** Self-contained CLI tool
- **Pre-processing:**
 - Irrelevant files skipped based on black- and whitelists
 - **Line number prefixing:** Helps LLMs maintain spatial awareness and prevent miscalculations
- Zero-Shot prompt, default model Claude Opus 4.5 (non-reasoning)
- **Output:** Human-readable text summaries + machine-readable JSON/CSV.

RQ3: How accurately can the developed LLM-based tool detect SQL antipatterns in Java code that uses jOOQ for database access?

$$P = 0.88, \quad R = 0.88, \quad F_1 = 0.88$$

- **Best performers:** *Implicit Columns* ($F_1 = 0.96$) and *31 Flavors* ($F_1 = 0.97$)
 - **Why?** Self-contained; specific detection patterns
- **Worst performers:** *Keyless Entry* ($F_1 = 0.69$), *Poor Man's Search Engine* ($F_1 = 0.68$), and *Fear of the Unknown* ($F_1 = 0.48$)
 - **Why?** Require cross-file context; imprecise localisation behaviour

RQ4: What patterns can be observed in the occurrence of SQL antipatterns in Java code that uses jOOQ for database access?

- 15,931 total antipatterns occurrences found in 602 projects, 26.5 per project
- **Dominant core:** "Implicit Columns" and "ID Required"
 - Contained by 89% and 87% of projects
- **Missing constraints:** "Keyless Entry" and "Beware of the Unknown"
 - Presence of one indicates the presence of other
- **Infection path:** Rarer SQL antipatterns are (weak) indicators of more common ones

RQ5: Which jOOQ API methods are most frequently associated with query antipattern occurrences?

Implicit Columns

- More than 75% associated with shorthands for fetching jOOQ records
 - e.g. `DSL.selectFrom(TABLE)`, `DSL.select().from(TABLE)`
- Only 12.9% explicitly show the intent to fetch all columns
 - e.g. `DSL.asterisk()`, `TABLE.fields()`

RQ5: Which jOOQ API methods are most frequently associated with query antipattern occurrences?

Poor Man's Search Engine

- More than 60% explicitly use `LIKE` and `ILIKE` operators with wildcards
 - `FIELD.like("%" + value + "%")`, `FIELD.ilike`
- Wildcards are added implicitly in 28% of cases
 - `FIELD.contains(value)`, `FIELD.containsIgnoreCase`

Highlights & Contributions

- Created an internally consistent ground truth
- First to treat LLM-based antipattern detection as a localisation task
- Developed the first available tool for detecting SQL antipatterns in jOOQ
- Performed a large-scale analysis of jOOQ projects

Thank you for listening!

1) Why have you not applied Abstract Syntax Trees at your SQL code classification method? How do you see the combination of LLMs and ASTs in SQL antipattern detection?

```
void fetchData() {  
    return dslContext.select(TABLE.asterisk()).from(TABLE).fetchOne();  
}
```

```
void fetchData(boolean fetchOther) {  
    List<Field> columns = getColumns(fetchOther);  
    return dslContext.select(columns).from(TABLE)  
        .innerJoin(OTHER).on(OTHER.ID.eq(TABLE.OTHER_ID))  
        .fetchOne();  
}
```

```
void getColumns(boolean fetchOther) {  
    List<Field> columns = new ArrayList<>();  
    columns.add(TABLE.ID);  
    if (fetchOther) columns.add(OTHER.asterisk());  
    return columns;  
}
```

```
public class UsersTable extends TableImpl<UsersTableRecord> {

    public final TableField<UsersTableRecord, String> GENDER =
        createField(DSL.name("gender"), SQLDataType.VARCHAR(255), this, "");

    @Override
    public List<Check<UsersTableRecord>> getChecks() {
        return Arrays.asList(
            Internal.createCheck(
                this,
                DSL.name("users_table_gender_check"),
                "(((gender)::text = ANY ((ARRAY['MALE'::character varying, 'FEMALE'::...]))))",
                true
            )
        );
    }
}
```

2) Why some existing no-LLM detection solutions like DBDeo were not included in related work comparison?

- Comparison only included static analysis methods
 - A. Performance of other methods was usually not evaluated
 - B. Evaluation was under vastly different conditions
- First party evaluation of DbDeo did not provide any comparison points
- Third party study only provided precision: $P = \frac{TP}{TP+FP}$

Antipattern	This study	DbDeo	SQLCheck
31 Flavors	0.97	0.52	0.99
Rounding Errors	1.00	0.99	0.98
...	...	...	...
Total	0.88	0.83	0.95

3) How would you overcome the size limitation of your original dataset? (Some antipattern types were not included in train-validate-test sets due to their rarity)

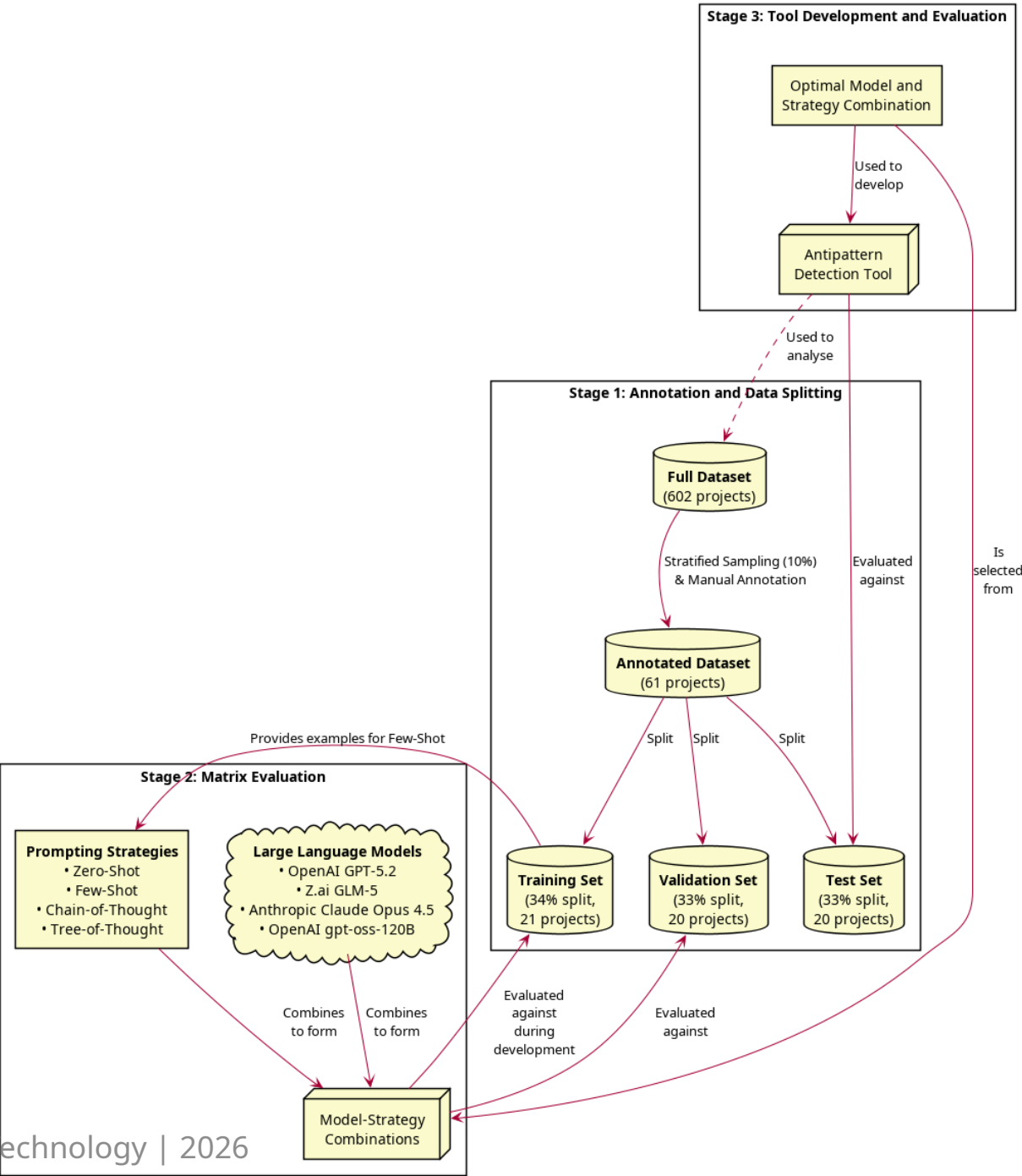
A. Complete shift of methodology

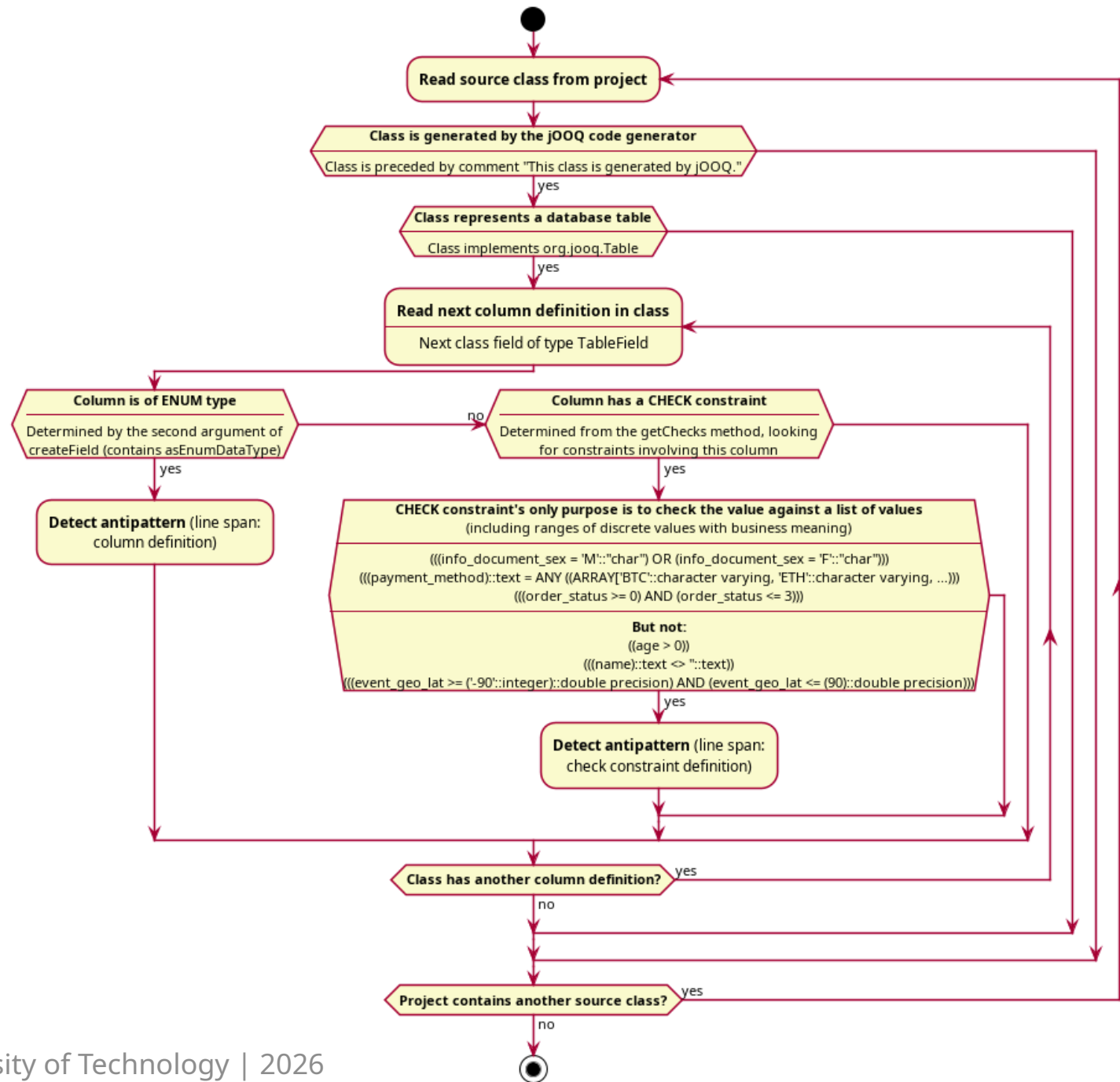
- Skip up-front annotation of ground truth
- Manually classify tool's detections as TP/FP
- Cache classifications, reuse for future iterations
- Replace recall ratio with absolute number of detections

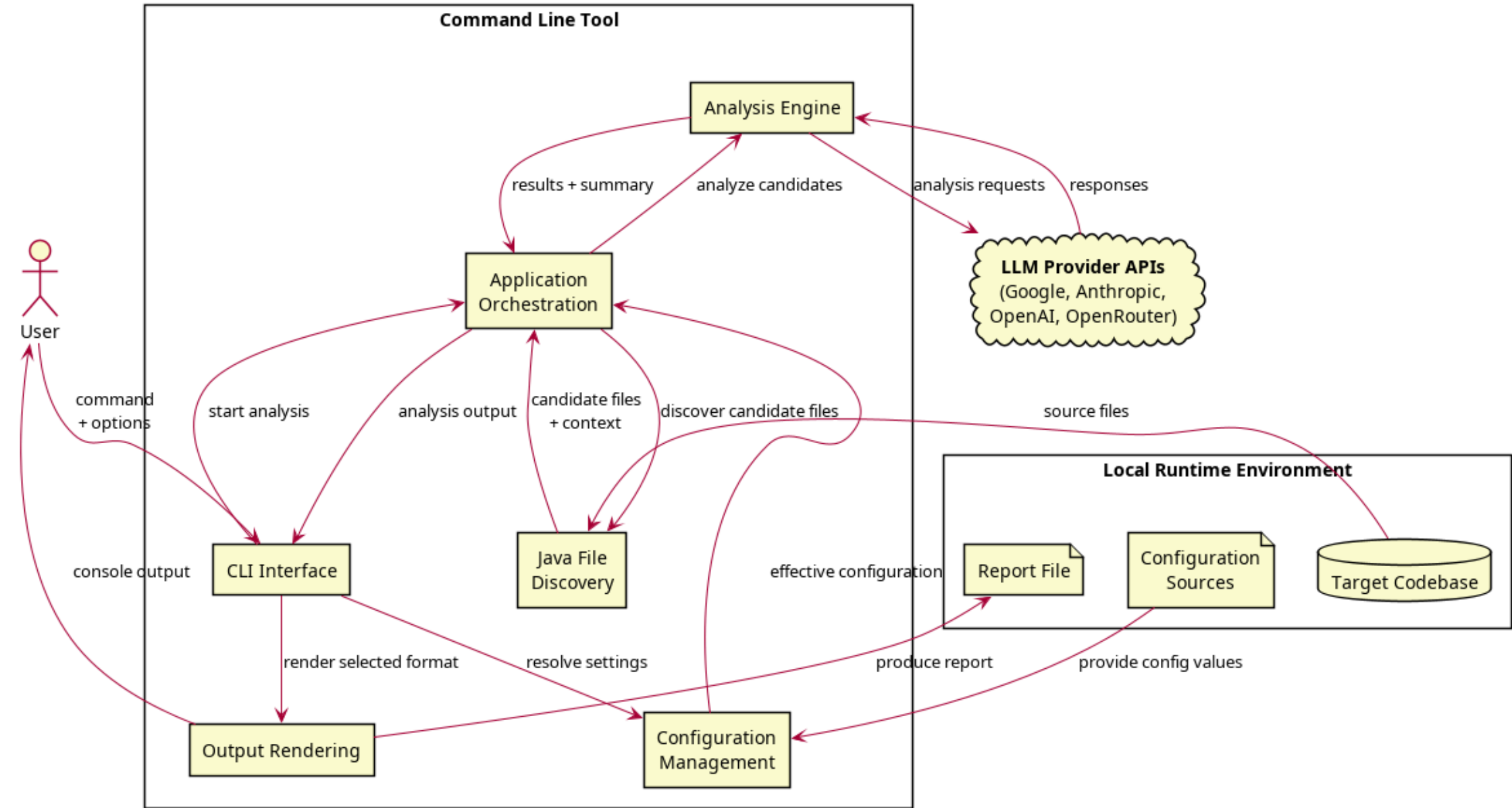
$$P = \frac{TP}{TP + FP}, \quad R = TP$$

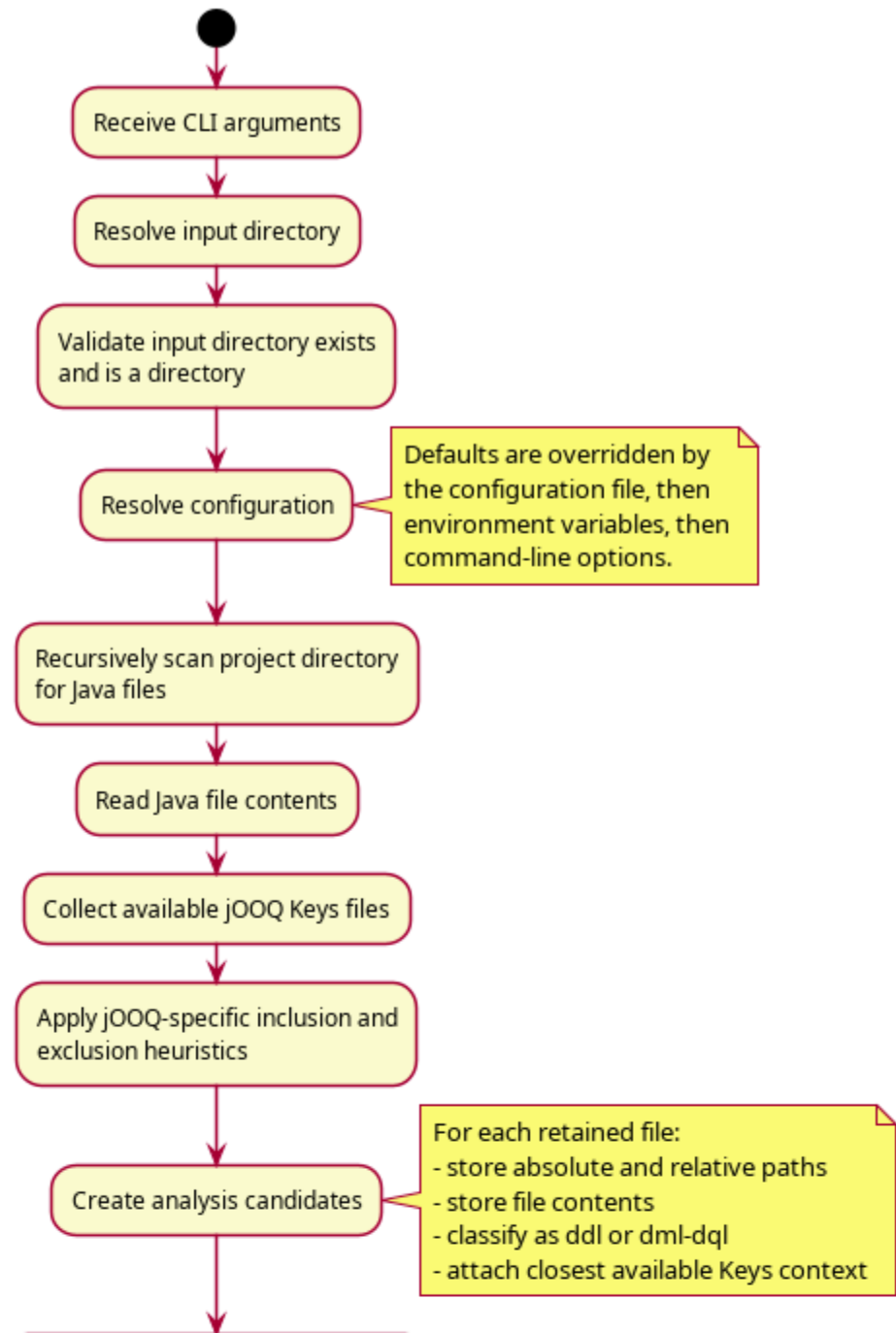
B. Augment data with synthetic samples

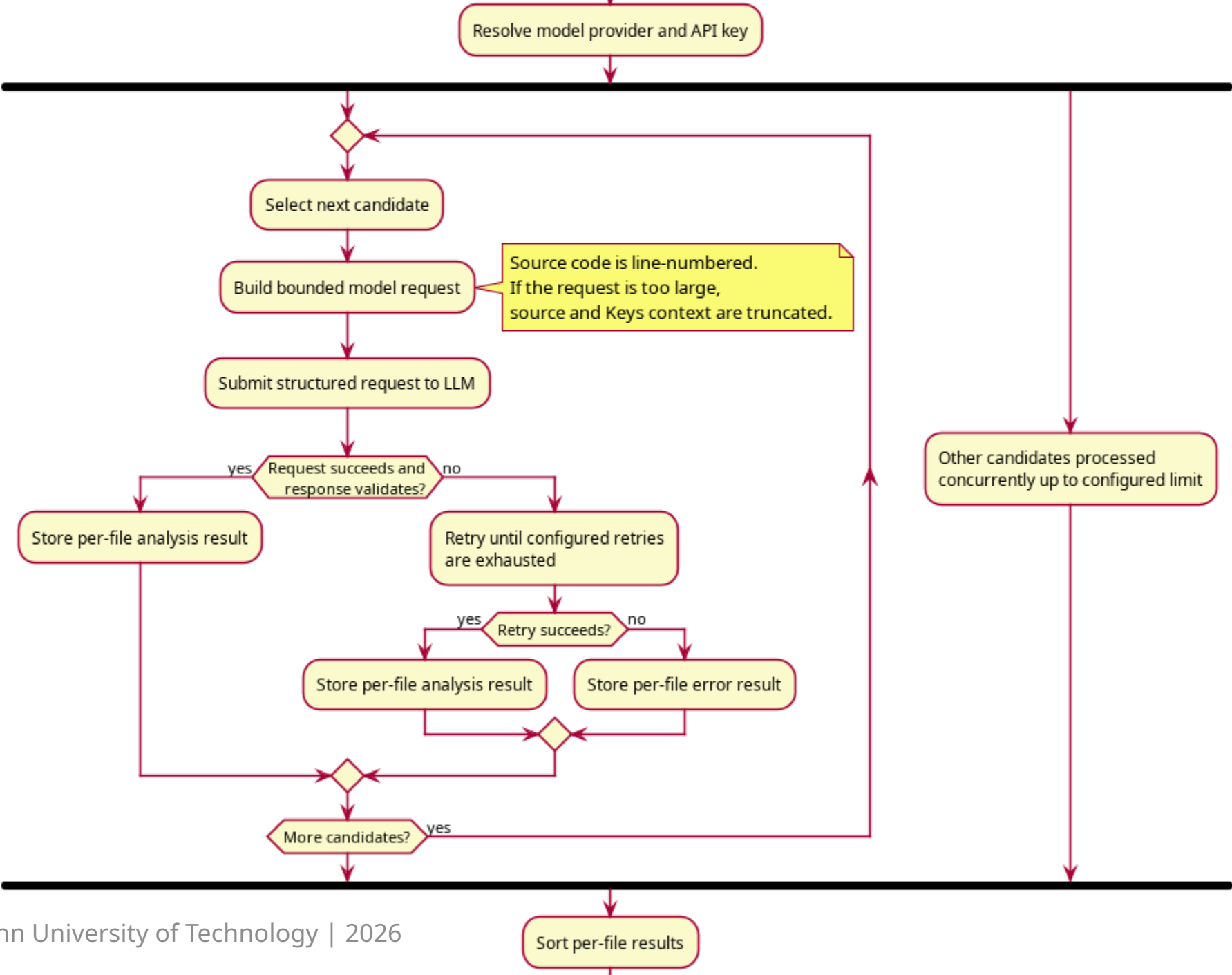


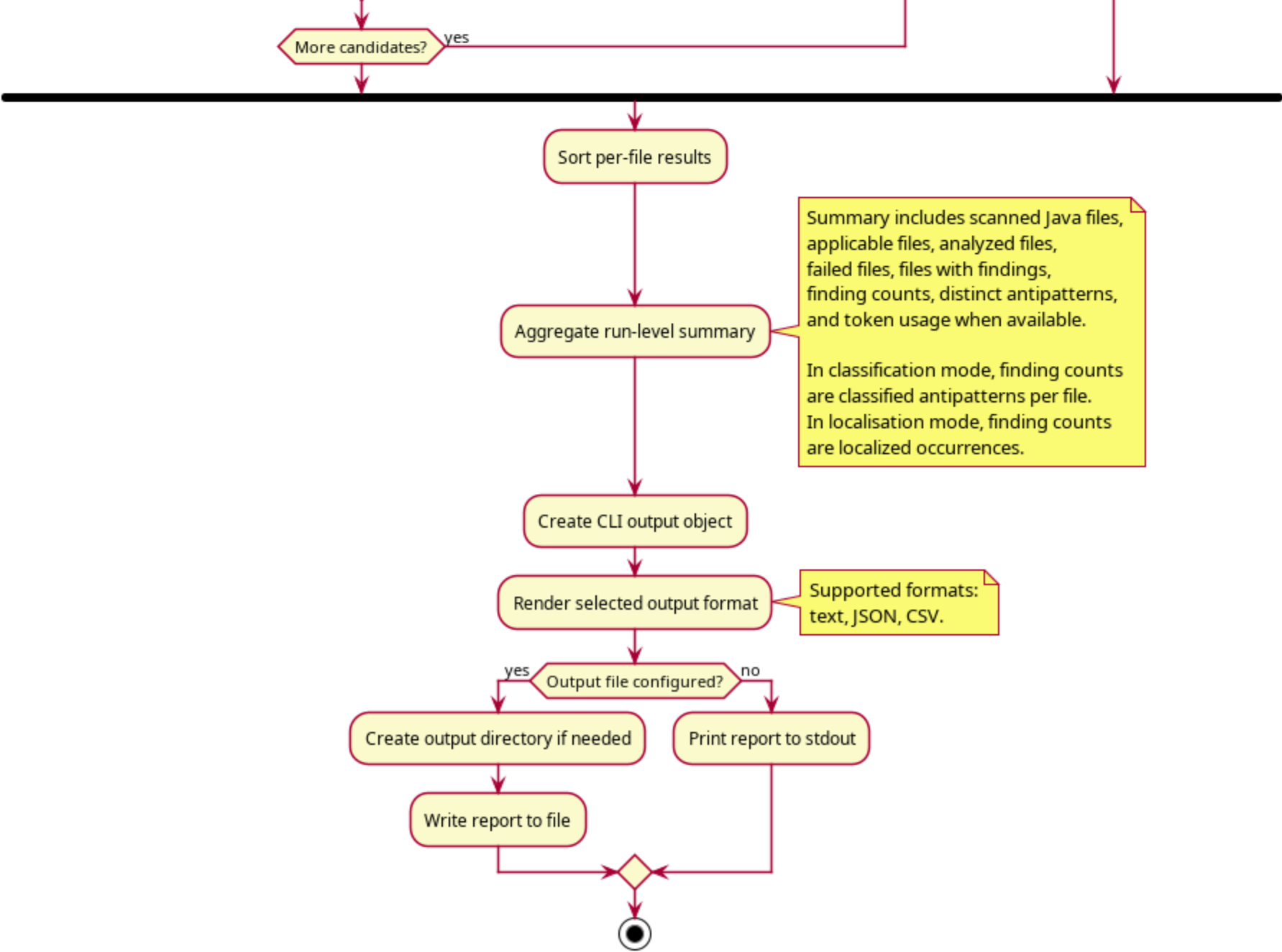












RQ3: How accurately can the developed LLM-based tool detect SQL antipatterns in Java code that uses jOOQ for database access?

Antipattern	Precision	Recall	F1-Score
Implicit Columns	0.98	0.94	0.96
ID Required	0.87	0.85	0.86
Keyless Entry	0.54	0.97	0.69
Fear of the Unknown	0.44	0.53	0.48
31 Flavors	0.97	0.97	0.97
Poor Man's Search Engine	0.76	0.62	0.68
Rounding Errors	1.00	0.81	0.90
Total	0.88	0.88	0.88